

Problem Statement

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Option: 2 (Implementing algorithms that are not in current literature)

Topic

We are exploring the topic of **Efficient Parallel Algorithms for Data Mining Operations**. The primary goal of this project is to develop and analyze efficient parallel algorithms for **data clustering** operations on large datasets. Data mining operations on large-scale data often require significant computational power, and by leveraging parallel computing techniques, we aim to optimize the performance of these operations.

Based on initial inputs from faculty members and related research, this topic aligns with our current interest. However, if we find that this topic is unsuitable for further development, we will shift our focus to **Parallel Graph Algorithms: Large-scale Graph Analytics**, specifically exploring algorithms related to **betweenness centrality** in large graphs.

Related Work

We studied few basic research papers and identified several techniques that can help us lay a foundation for the project. [1] provided an overview of efficient data mining techniques that help in managing large datasets, offering useful guidelines for structuring and optimizing our project. [2] demonstrated how parallel algorithms can speed up data clustering by distributing tasks across multiple processors, a strategy we aim to apply to graph problems. Additionally, Han et al.

For the second topic, that is betweenness centrality, we looked up the following references: [4] proposed a method for estimating and updating betweenness centrality using progressive algorithms and shortest paths approximation. [5] introduced techniques for approximating betweenness centrality by minimizing the number of shortest path computations required, which offers a practical way to speed up the process in parallel environments. Lumsdaine et al. [3] discussed the challenges of scaling graph algorithms in parallel computing, particularly in managing load balance and communication between processors. Their insights will help us design more efficient algorithms. Zaki et al.

References

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- [4] N. Xiang, Q. Wang, and M. You, "Estimation and update of betweenness centrality with progressive algorithm and shortest paths approximation," *Scientific Reports*, vol. 13, article 17110, 2023.

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- [6] C. C. Aggarwal and C. K. Reddy, *Data Clustering: Algorithms and Applications*, 1st ed., Chapman and Hall/CRC, 2014, doi:10.1201/9781315373515.